

DATA ANALYST

Intenship project 1

DESCRIPTION

The objective of this project is to analyze sales data using SQL for data extraction, Python (Google Colab) for data processing, and Microsoft Power BI for visualization to derive business insights.

PREPARED BY

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(21-02-2026)

MY WORK

In this project, I performed end-to-end sales data analysis using SQL, Python (Google Colab), and Microsoft Power BI. I wrote SQL queries to extract insights, used Python for data processing and calculations, and built an interactive dashboard to visualize revenue, profit, orders, and regional performance for business decision-making.

DATASET USED

E-commerce data

FINAL DATASET

Final E-commerce data

SQL QUERIES

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Database Structure Browse Data Edit Pragma Execute SQL

SQL 1*

```
1 SELECT * FROM ecommerce_data LIMIT 10;
```

	customer_id	customer_name	city	region	order_id	order_date	product_id	product_name	category	quantity	price
1	1	Aarav Sharma	Delhi	North	1001	1/1/2024	P01	Wireless Mouse	Electronics	1	799
2	2	Diya Patel	Ahmedabad	West	1002	1/2/2024	P02	Bluetooth Headphones	Electronics	2	1999
3	3	Rohan Verma	Mumbai	West	1003	1/3/2024	P03	Notebook	Stationery	3	60
4	4	Ananya Iyer	Chennai	South	1004	1/4/2024	P04	Water Bottle	Home	4	299
5	5	Kabir Singh	Delhi	North	1005	1/5/2024	P05	Office Chair	Furniture	5	7499
6	6	Meera Nair	Kochi	South	1006	1/6/2024	P06	Desk Lamp	Home	1	899
7	7	Airjun Mehta	Pune	West	1007	1/7/2024	P07	Backpack	Accessories	2	2499
8	8	Priya Das	Kolkata	East	1008	1/8/2024	P08	Pen Set	Stationery	3	120
9	9	Sahil Khan	Delhi	North	1009	1/9/2024	P09	Smart Watch	Electronics	4	5999
10	10	Neha Gupta	Jaipur	North	1010	1/10/2024	P10	Coffee Mug	Home	5	349

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Database Structure Browse Data Edit Pragma Execute SQL

SQL 1*

```
1 SELECT * FROM ecommerce_data LIMIT 10;  
2 SELECT COUNT(DISTINCT order_id) AS total_orders  
3 FROM ecommerce_data;
```

	total_orders
1	200

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Database Structure Browse Data Edit Pragma Execute SQL

SQL 1*

```
1 SELECT * FROM ecommerce_data LIMIT 10;  
2 SELECT COUNT(DISTINCT order_id) AS total_orders  
3 FROM ecommerce_data;  
4 SELECT SUM(total_sales) AS total_revenue  
5 FROM ecommerce_data;
```

	total_revenue
1	1519320

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Database Structure Browse Data Edit Pragma Execute SQL

SQL 1*

```
1 SELECT * FROM ecommerce_data LIMIT 10;  
2 SELECT COUNT(DISTINCT order_id) AS total_orders  
3 FROM ecommerce_data;  
4 SELECT SUM(total_sales) AS total_revenue  
5 FROM ecommerce_data;  
6 SELECT SUM(quantity) AS total_quantity  
7 FROM ecommerce_data;
```

	total_quantity
1	600

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Database Structure Browse Data Edit Pragma Execute SQL

SQL 1*

```
2 SELECT COUNT(DISTINCT order_id) AS total_orders  
3 FROM ecommerce_data;  
4 SELECT SUM(total_sales) AS total_revenue  
5 FROM ecommerce_data;  
6 SELECT SUM(quantity) AS total_quantity  
7 FROM ecommerce_data;  
8 SELECT category,  
9 SUM(total_sales) AS revenue  
10 FROM ecommerce_data  
11 GROUP BY category  
12 ORDER BY revenue DESC;
```

	category	revenue
1	Furniture	749900
2	Electronics	575860
3	Accessories	99960
4	Home	76800
5	Stationery	10800

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Database Structure Browse Data Edit Pragma Execute SQL

SQL 1*

```
7 FROM ecommerce_data;  
8 SELECT category,  
9 SUM(total_sales) AS revenue  
10 FROM ecommerce_data  
11 GROUP BY category  
12 ORDER BY revenue DESC;  
13 SELECT category,  
14 SUM(quantity) AS total_quantity  
15 FROM ecommerce_data  
16 GROUP BY category  
17 ORDER BY total_quantity DESC;
```

	category	total_quantity
1	Home	200
2	Electronics	140
3	Stationery	120
4	Furniture	100
5	Accessories	40

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Database Structure Browse Data Edit Pragmas Execute SQL

```

12 ORDER BY revenue DESC;
13 SELECT category,
14     SUM(quantity) AS total_quantity
15 FROM ecommerce_data
16 GROUP BY category
17 ORDER BY total_quantity DESC;
18 SELECT region,
19     SUM(total_sales) AS revenue
20 FROM ecommerce_data
21 GROUP BY region
22 ORDER BY revenue DESC;
  
```

region	revenue
1 North	1280700
2 West	183520
3 South	41900
4 East	7200

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Database Structure Browse Data Edit Pragmas Execute SQL

```

18 SELECT region,
19     SUM(total_sales) AS revenue
20 FROM ecommerce_data
21 GROUP BY region
22 ORDER BY revenue DESC;
23 SELECT
24     strftime('%m', order_date) AS month,
25     SUM(total_sales) AS revenue
26 FROM ecommerce_data
27 GROUP BY month
28 ORDER BY month;
  
```

month	revenue
1 NULL	1513320

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Database Structure Browse Data Edit Pragmas Execute SQL

```

24     strftime('%m', order_date) AS month,
25     SUM(total_sales) AS revenue
26 FROM ecommerce_data
27 GROUP BY month
28 ORDER BY month;
29 SELECT product_name,
30     SUM(total_sales) AS revenue
31 FROM ecommerce_data
32 GROUP BY product_name
33 ORDER BY revenue DESC
34 LIMIT 5;
  
```

product_name	revenue
1 Office Chair	749900
2 Smart Watch	479920
3 Backpack	99960
4 Bluetooth Headphones	79960
5 Coffee Mug	34900

In this project, I used nine SQL queries to extract meaningful insights from the sales dataset. The queries included calculating total revenue, total profit, total quantity sold, and total number of orders. I also analyzed revenue and profit by category and region to identify high-performing segments. Additional queries were used to determine the top 10 products based on revenue and to examine sales trends over time. Each query helped transform raw data into structured information for analysis. These SQL queries formed the foundation for further processing in Python and visualization in Microsoft Power BI, enabling clear business insights and performance evaluation.

SALES DATA ANALYSIS USING PYTHON IN GOOGLE COLAB

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen

Upload widget is only available when the cell has been executed

in the current browser session. Please rerun this cell to enable.

Saving ecommerce_cleaned.csv to ecommerce_cleaned.csv

```
import pandas as pd

df = pd.read_csv("ecommerce_cleaned.csv")

df.head()
```

	customer_id	customer_name	city	region	order_id	order_date	product_id	product_name
0	1	Aarav Sharma	Delhi	North	1001	1/1/2024	P01	Wireless Mouse
1	2	Diya Patel	Ahmedabad	West	1002	1/2/2024	P02	Bluetooth Headphones
2	3	Rohan Verma	Mumbai	West	1003	1/3/2024	P03	Notebook
3	4	Ananya Iyer	Chennai	South	1004	1/4/2024	P04	Water Bottle
4	5	Kabir Singh	Delhi	North	1005	1/5/2024	P05	Office Chair

```
df.corr(numeric_only=True)
```

	customer_id	order_id	quantity	price	total_sales
customer_id	1.000000	0.049750	0.49236	0.169501	0.179180
order_id	0.049750	1.000000	6	0.008433	0.008914
quantity	0.492366	0.024495	0.02449	0.400334	0.534873
price	0.169501	0.008433	5	1.000000	0.974251
total_sales	0.179180	0.008914	1.00000	0.974251	1.000000

0

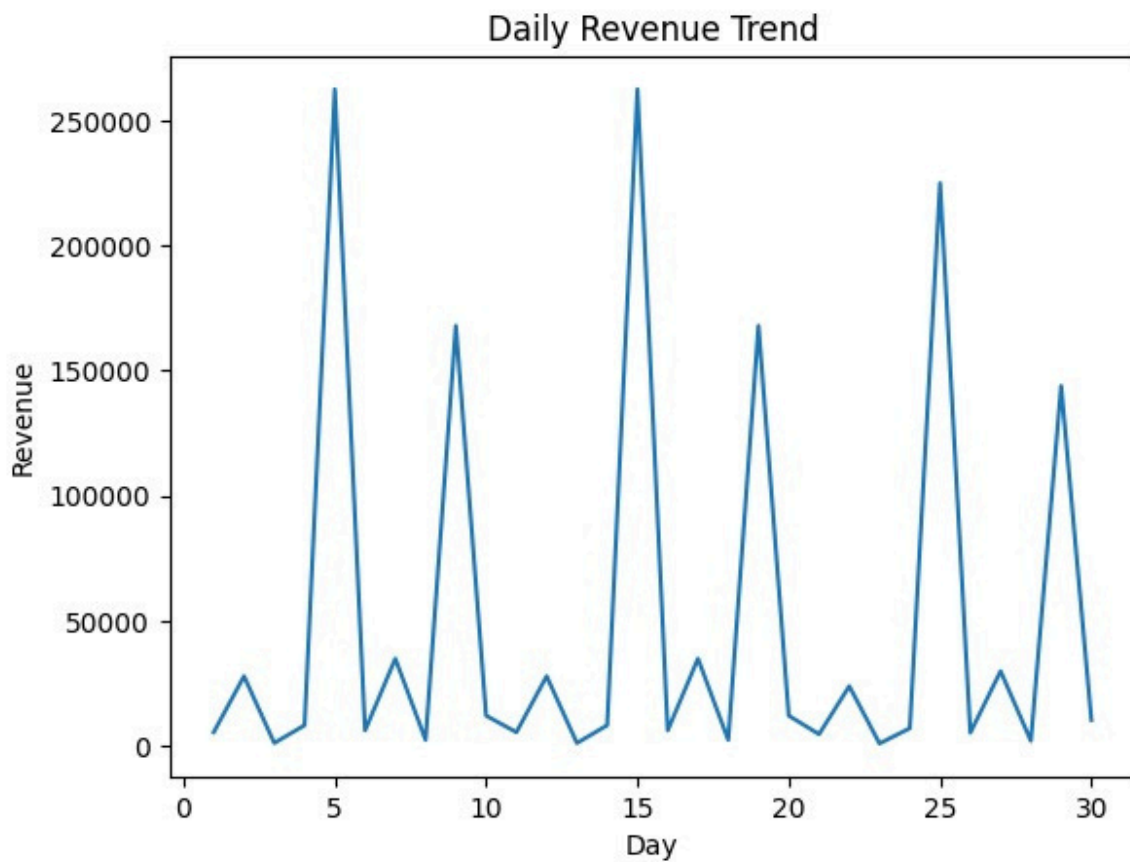
```
df['quantity'].corr(df['total_sales'])
```

```
np.float64(0.5348729153986913)
```

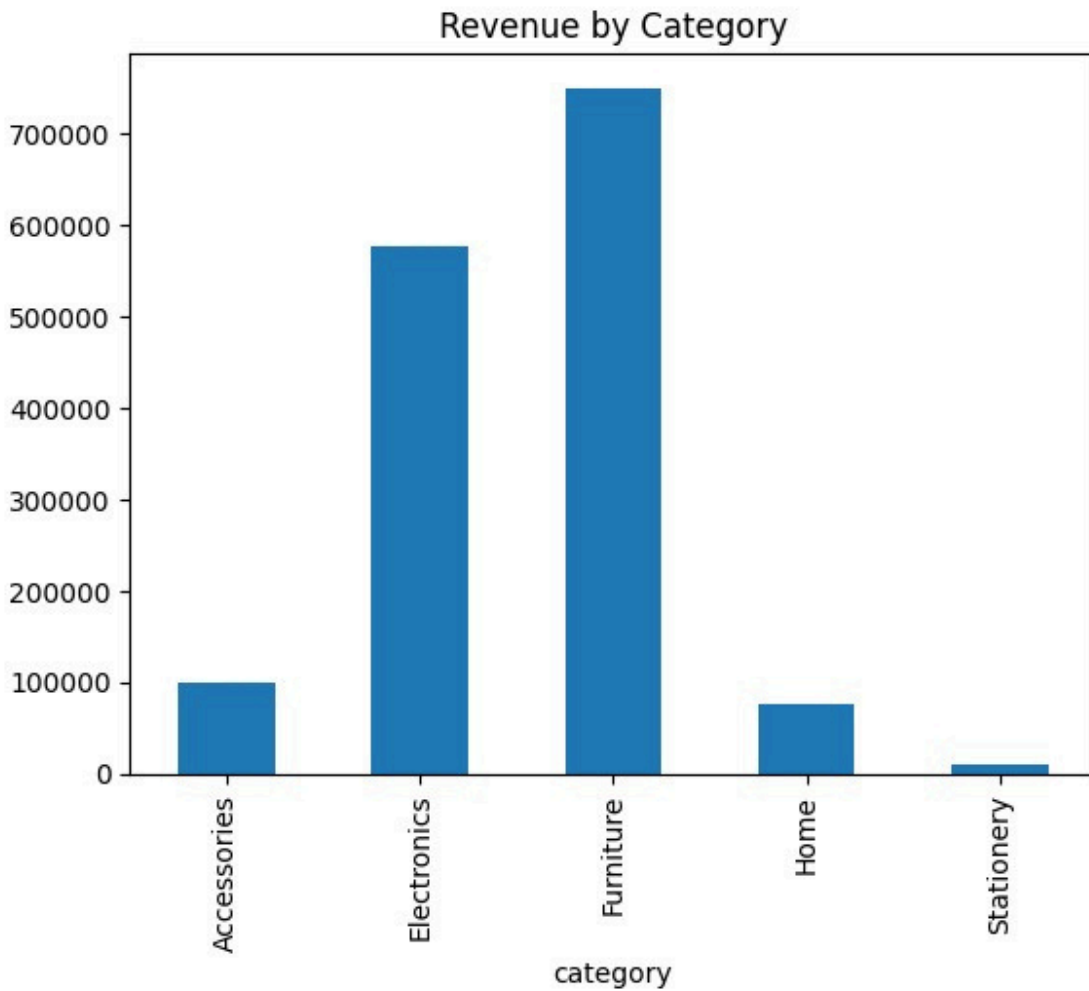
```
df['day'] = df['order_date'].dt.day
daily = df.groupby('day')['total_sales'].sum()

daily.plot(kind='line')
plt.title("Daily Revenue Trend")
plt.xlabel("Day")
```

```
plt.ylabel("Revenue")  
plt.show()
```

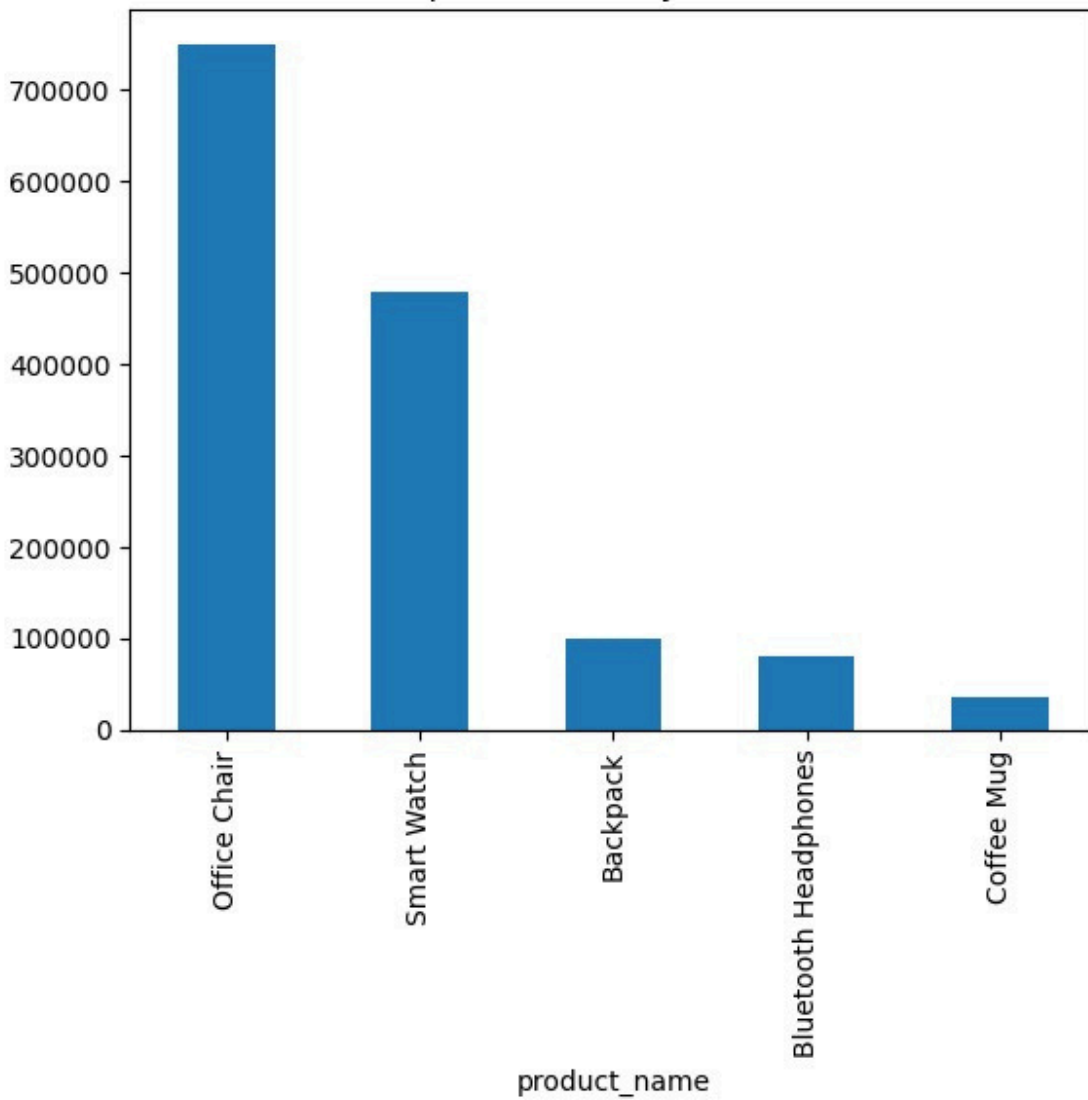


```
category_sales = df.groupby('category')['total_sales'].sum()  
  
category_sales.plot(kind='bar')  
plt.title("Revenue by Category")  
plt.show()
```



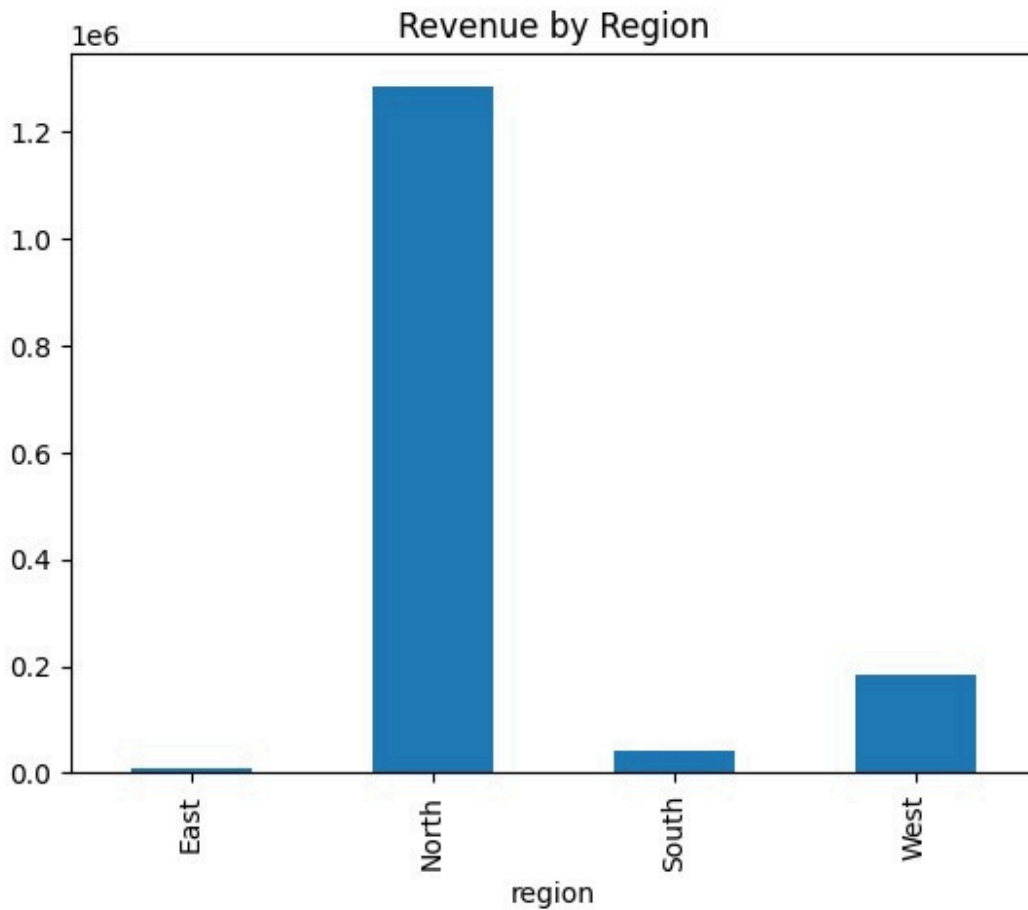
```
top_products = df.groupby('product_name')['total_sales'].sum().sort_values(ascending=False).head(5)
top_products.plot(kind='bar')
plt.title("Top 5 Products by Revenue")
plt.show()
```

Top 5 Products by Revenue



```
region_sales = df.groupby('region')['total_sales'].sum()

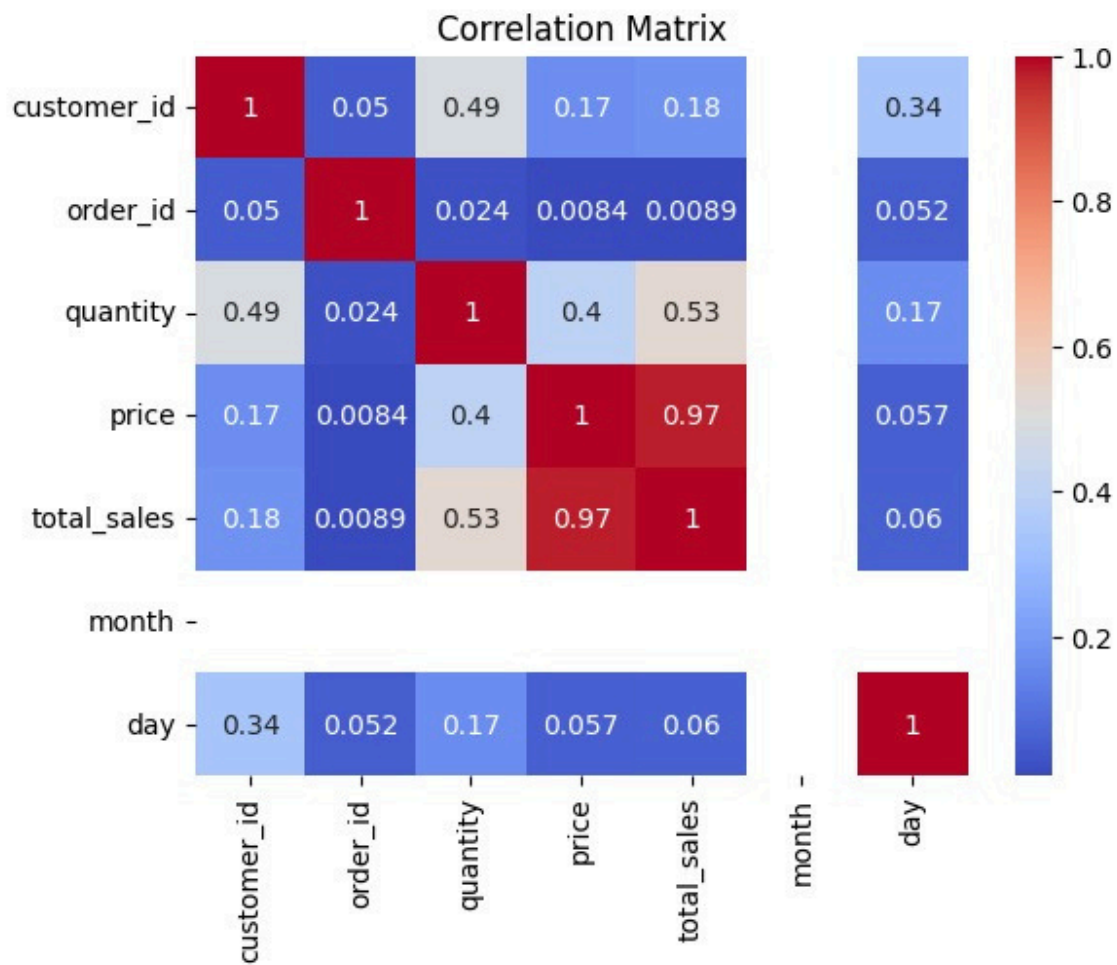
region_sales.plot(kind='bar')
plt.title("Revenue by Region")
plt.show()
```



```
import seaborn as sns

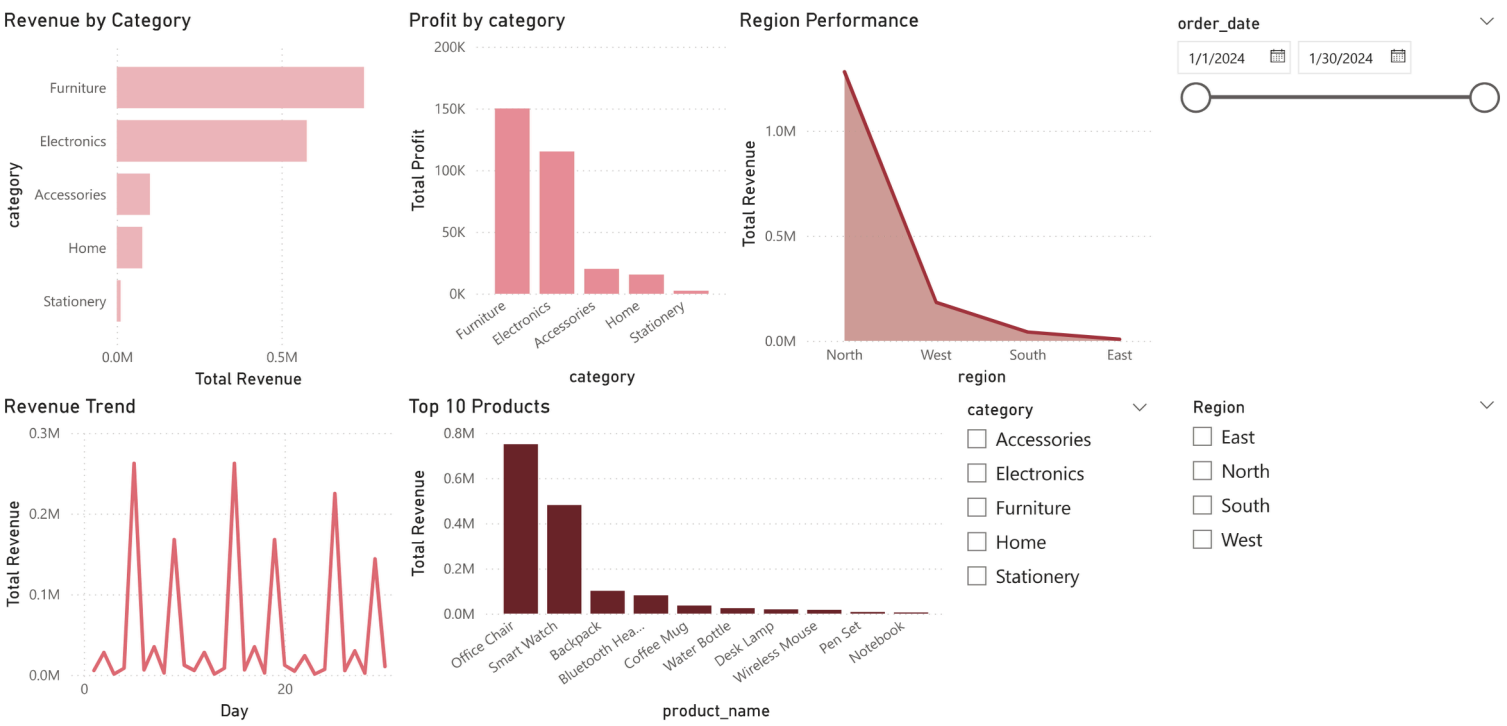
corr = df.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```



```
df.to_csv("final_ecommerce_data.csv", index=False)
```

Sales Performance Dashboard



The Sales Performance Dashboard provides an interactive analysis of revenue, profit, orders, and quantity using Microsoft Power BI. It includes key performance indicators such as Total Revenue, Total Profit, Profit Margin, Total Orders, and Total Quantity to give a quick overview of business performance. The dashboard visualizes revenue and profit by category and region, highlights the top 10 products, and shows sales trends over time. Interactive slicers for category, region, and order date allow users to filter and explore the data dynamically. This dashboard helps in identifying high-performing segments, understanding trends, and supporting data-driven business decisions